INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II , Fourth Semester Examination, March - 2025

Subject Code: BCACC401
Subject Name: Database Management Systems

Full Marks: 30 Set No: 1 Duration: 1 Hour

Part - A
Answer any 5 out of 6 questions
Each question carries 2 marks (2 × 5)

1. Define Functional Dependency. **2**

2. Define Normalization. **2**

3. Define Database Transaction. **2**

4. What is data redundancy? **2**

5. Define the concept of primary key. **2**

6. What is the purpose of where clause in SQL. **2**

Part - B
Answer any 2 out of 3 questions
Each question carries 5 marks (5 × 2)

7. Differentiate between 3NF and BCNF. **5**

8. What is Partial dependency? **5**

9. Define ACID property of Transaction. **5**

Part - C
Answer any 1 out of 2 questions
Each question carries 10 marks (10 × 1)

10. Explain the different deadlock handling techniques in DBMS. **10**

11. R(A,B,C,D,E,F,G,H,I,J) with given set of FD **10**
{AB→C , A→DE, B→F, F→GH, D→IJ}
a) Find the Candidate key of R
b) Decompose The relation into 3NF
c) Show the primary key and Foreign for each sub relation after decomposing R into 3NF.

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Subject Code: BCACC401
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Full Marks: 30 Set No: **2** Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. What is indexing in DBMS? Name two types of indexing used in DBMS. **2**

2. How does a dense index differ from a sparse index? **2**

3. What is the role of foreign keys in relational database design? **2**

4. What is normalization in database design? **2**

5. Define a clustered index. **2**

6. Define the term "multilevel indexing." **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Write the difference between primary and secondary Indexing. **5**

8. Explain the lost update problem with proper schedule examples. **5**

9. Define ACID property of Transaction. **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Explain in detail the steps involved in designing an ER Model for a university Management system. **10**

11. In a relation R(A,B,C,D,E) and the FDs are A→BC,CD→E,B→D,E→A Find the candidate key **10**

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Subject Code: BCACC401
Subject Name: Database Management Systems

Full Marks: 30 Set No: 3 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. What is indexing in DBMS? Name two types of indexing used in DBMS. **2**

2. Why do we need concurrency control mechanisms? **2**

3. What is the role of foreign keys in relational database design? **2**

4. What is normalization in database design? **2**

5. Define a clustered index. **2**

6. Mention the two conditions of 2 Phase Locking. **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Write the difference between Where and Having Clause. **5**

8. Explain the incorrect summary problem with proper schedule examples. **5**

9. Define ACID property of Transaction. **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Explain in detail the steps involved in designing an ER Model for a university Management system. **10**

11. Evaluate whether the following schedule is conflict serializable or not. **10**
S : $R_1(A)$, $R_2(A)$, $R_1(B)$, $R_2(B)$, $R_3(B)$, $W_1(A)$, $W_2(B)$.
Design the Transaction Table and Precedence Graph to analyse the result.

--End--